# Yifei Zhang

Applied Scientist, Amazon Alexa AI – Boston, Massachusetts
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## Education

**Emory University** **Atlanta, GA**
*Ph.D. in Computer Science* *Aug. 2022 – Mar. 2026*
Advisor: Prof. Liang Zhao
Dissertation: Benchmarking, Correcting, and Accelerating AI Reasoning across Diverse Modalities

**Columbia University** **New York, NY**
*M.S. in Data Science* *Sept. 2020 – Dec. 2021*

**Duke University** **Durham, NC**
*Master of Engineering Management* *Aug. 2019 – May 2020*

**Dalian University of Technology** **Dalian, China**
*B.E. in Engineering Mechanics* *Sept. 2015 – June 2019*

## Research Focus

AI agents and evaluation; explainable and trustworthy AI; multimodal reasoning.

Develop benchmarks and methods for long-horizon dialogue, memory, visual explanation, retrieval-augmented generation, and reasoning across language, vision, and clinical data. Representative work includes ATOD, Hear2Act, MAGI, ELAD, and Saliency-Bench.

## Research and Industry Experience

**Amazon** **Boston, MA**
*Applied Scientist* *Apr. 2026 – Present*
- Develop scalable evaluation methods for agentic conversational systems, covering reliability, task completion, conversational quality, and multimodal behavior.
- Combine evaluation signals into interpretable assessments for model and system improvement.
- Research privacy-aware model adaptation and distillation under practical system constraints.

**Amazon** **Boston, MA**
*Applied Scientist Intern* *May 2025 – Aug. 2025*
- Developed a memory system and multi-agent evaluation framework for conversational systems.
- Built synthetic dialogue-generation and benchmarking pipelines for long-horizon, multi-turn, and concurrent user tasks.

**Amazon AGI** **Boston, MA**
*Research Scientist Intern* *June 2024 – Sept. 2024*
- Developed reward modeling and preference-based evaluation workflows for LLM-driven conversational interactions.
- Built synthetic preference-data generation and parameter-efficient fine-tuning pipelines.

**Emory University** **Atlanta, GA**
*Ph.D. Student Researcher* *Aug. 2022 – Mar. 2026*
Developed evaluation benchmarks and explanation-guided methods for reliable AI across language, vision, and multimodal data.

## Publications

Full publication list: Google Scholar.

### Peer-Reviewed and Accepted Papers

**Dual Decomposition for Text-to-Image Retrieval and Generation**
M. Zhu, S. Cheng, G. Bai, **Yifei Zhang**, and L. Zhao
*The 20th Conference of the European Chapter of the Association for Computational Linguistics (EACL), Findings, 2027 (accepted).*

**Beyond Efficiency: A Systematic Survey of Resource-Efficient Large Language Models**
G. Bai, Z. Chai, C. Ling, S. Wang, J. Lu, N. Zhang, T. Shi, Z. Yu, M. Zhu, **Yifei Zhang**, X. Song, C. Yang, Y. Cheng, and L. Zhao
*ACM Computing Surveys (CSUR), 2026.*

**ATOD: An Evaluation Framework and Benchmark for Agentic Task-Oriented Dialogue Systems**
**Yifei Zhang**, H. Nayyeri, R. Khaziev, E. Yilmaz, G. Tur, D. Hakkani-Tür, and H. P. Thadakamalla
*The 15th International Joint Conference on Natural Language Processing and the 5th Conference of the Asia-Pacific Chapter of the Association for Computational Linguistics (AACL-IJCNLP), 2026.*

**Hear2Act: Benchmarking When Prosody Should Change What an Assistant Does**
X. Liu, H. Nayyeri, D. Hakkani-Tür, E. Yilmaz, J.-K. Kim, **Yifei Zhang**, C. Peris, and H. Thadakamalla
*Conference on Neural Information Processing Systems (NeurIPS), RTCA Workshop, 2026 (accepted poster).*

**Saliency-Bench: A Comprehensive Benchmark for Evaluating Visual Explanations**
**Yifei Zhang**, J. Song, S. Gu, T. Jiang, B. Pan, G. Bai, and L. Zhao
*The 31st ACM SIGKDD Conference on Knowledge Discovery and Data Mining (KDD), Datasets and Benchmarks Track, 2025.*

**GraphNarrator: Generating Textual Explanations for Graph Neural Networks**
B. Pan, Z. Xiong, G. Wu, Z. Zhang, Y. Hu, **Yifei Zhang**, and L. Zhao
*The 63rd Annual Meeting of the Association for Computational Linguistics (ACL), 2025.*

**Distilling Large Language Models for Text-Attributed Graph Learning**
B. Pan, Z. Zhang, **Yifei Zhang**, Y. Hu, and L. Zhao
*The 33rd ACM International Conference on Information and Knowledge Management (CIKM), 2024.*

**Multi-Prompt Fine-Tuning of Foundation Models for Enhanced Biomedical Image Segmentation**
X. Li, **Yifei Zhang**, and L. Zhao
*AI for Health Equity and Fairness, Studies in Computational Intelligence, vol. 1164, pp. 83–91, Springer, 2024.*

**DUE: Dynamic Uncertainty-Aware Explanation Supervision via 3D Imputation**
Q. Zhao, **Yifei Zhang**, M. Zhu, S. Gu, Y. Gao, X. Yang, and L. Zhao
*The 30th ACM SIGKDD Conference on Knowledge Discovery and Data Mining (KDD), 2024.*

**ELAD: Explanation-Guided Large Language Models Active Distillation**
**Yifei Zhang**, B. Pan, C. Ling, Y. Hu, and L. Zhao
*The 62nd Annual Meeting of the Association for Computational Linguistics (ACL), Findings, 2024.*

**Visual Attention Prompted Prediction and Learning**
**Yifei Zhang**, B. Pan, S. Gu, G. Bai, M. Qiu, X. Yang, and L. Zhao
*The 33rd International Joint Conference on Artificial Intelligence (IJCAI), 2024.*

**STES: A Spatiotemporal Explanation Supervision Framework**
D. Yu, B. Chen, Y. Li, S. Dhakal, **Yifei Zhang**, Z. Liu, M. Zhang, J. Zhang, and L. Zhao
*SIAM International Conference on Data Mining (SDM), 2024.*

**MAGI: Multi-Annotated Explanation-Guided Learning**
**Yifei Zhang**, S. Gu, Y. Gao, B. Pan, X. Yang, and L. Zhao
*IEEE/CVF International Conference on Computer Vision (ICCV), 2023.*

**ESSA: Explanation Iterative Supervision via Saliency-Guided Data Augmentation**
S. Gu, **Yifei Zhang**, Y. Gao, X. Yang, and L. Zhao
*The 29th ACM SIGKDD Conference on Knowledge Discovery and Data Mining (KDD), 2023.*

## Preprints

**Agentic Commerce: A Survey of How AI Agents Are Reshaping Commerce**
**Yifei Zhang**, B. Pan, M. Zhu, J. Pei, and L. Zhao
*TechRxiv preprint, 2026.*

**EmoLLM: Appraisal-Grounded Cognitive-Emotional Co-Reasoning in Large Language Models**
**Yifei Zhang**, M. Li, H. Gao, and L. Zhao
*arXiv:2603.16553, 2026.*

**Explainable Cross-Disease Reasoning for Cardiovascular Risk Assessment from Low-Dose Computed Tomography**
**Yifei Zhang**, J. Zhang, M. Safari, X. Yang, and L. Zhao
*arXiv:2511.06625, 2025.*

**MEGL: Multimodal Explanation-Guided Learning**
**Yifei Zhang**, T. Jiang, B. Pan, J. Wang, G. Bai, and L. Zhao
*arXiv:2411.13053, 2024.*

**Saliency-Guided Hidden Associative Replay for Continual Learning**
G. Bai, Q. Zhao, X. Jiang, **Yifei Zhang**, and L. Zhao
*arXiv:2310.04334, 2023.*

**SurroCBM: Concept Bottleneck Surrogate Models for Generative Post-hoc Explanation**
B. Pan, Z. Liu, **Yifei Zhang**, and L. Zhao
*arXiv:2310.07698, 2023.*

**Controllable Data Generation Via Iterative Data-Property Mutual Mappings**
B. Pan, M. Qin, S. Wang, **Yifei Zhang**, and L. Zhao
*arXiv:2310.07683, 2023.*

## Academic Service

**Journal Reviewer:** ACM TIST; IEEE TBD; IEEE TNNLS; JACMP; Expert Systems with Applications; Information Processing and Management; Neurocomputing; Information Systems Journal.
**Conference Reviewer / Program Committee:** ICLR 2025–2027; NeurIPS 2026; AAAI 2027; CVPR 2025–2026; ICCV 2025; ECCV 2026; BMVC 2026; ACL ARR 2025–2026; KDD Research Track 2025–2026; KDD Datasets and Benchmarks Track 2025–2026; MemAgents Workshop (ICLR 2026).
**Workshop Organization:** Publicity Chair, DeepSpatial 2024 (KDD); Student Co-organizer, FM4sBio 2025 (AAAI).

## Honors, Grants, and Memberships

- Sigma Xi, Full Membership.
- Tinker Research Grant.
- KDD 2023 Student Travel Award.
- Dalian University of Technology Overseas Alumni Association Seagle Scholarship Award.
- Academic Excellence Scholarship Award, Dalian University of Technology.
- Technological Innovation Scholarship Award, Dalian University of Technology.

## Technical Skills

**Programming:** Python, C/C++, MATLAB, Java, Shell
**AI / LLM Frameworks:** PyTorch, Transformers, TensorFlow, vLLM, verl, TRL, PEFT
**Training and Infrastructure:** Accelerate, DeepSpeed, AWS, Docker, Linux, Git